U19CST69	SOFTWARE TESTING	L	T	P	C
	(Common to CSE & IT)	3	0	0	3

Pre-Requisites : Software Engineering Concepts, Programming using Java

Objectives:

- To learn software testing life cycle and its process.
- Understand the process of static testing and test design techniques.
- Know in details about risk and defect management techniques.
- Understand the importance of test data Management in Excel.
- To learn practically about automated tools used for software testing.

Course Outcomes:

At the end of this course students will demonstrate the ability to

CO1 : Understand complete software testing life cycle and Demonstrate understanding of various terms and technologies used in testing domain.

CO2 : Demonstrate understanding of generating test plan and designing test cases.

CO3 : Demonstrate understanding of defect management life cycle and ability to use an open source tool for defect management.

CO4 : Demonstrate understanding of test management process. Given a business scenario, identify and write the test plan, design test cases, document test cases using an open source test management tool.

CO5 : Demonstrate understanding of automation testing. Given an web application for testing, create test script and execute automated tests using an open source automation testing tool.

Unit I Introduction To Testing – Software Testing Life cycle 9

Introduction to Testing – Need of Testing –Testing Principle- Role of Tester, Psychology of Testing and Quality-Verification and validation. Overview of STLC-V-Model- SDLC vs STLC, different stages in STLC, document templates generated in different phases of STLC, Levels of testing –Component testing –integration testing-System testing-acceptance testing- Test types-functional –nonfunctional-structural testing

Unit II Static Testing and Test Design Techniques 9

Static testing and dynamic testing, reviews-roles and responsibilities –types of review walkthroughs-Technical review-Inspection –Static analysis tools. Various test categories, test design techniques for different categories of tests-White box –black box-Experienced based testing Designing test cases using MS-Excel.

Unit III Test and Defect Management 9

Test plan –Estimate and Strategies-Risk-level of risk –product risk –project risk-Incident Management. Documenting test plan and test case, effort estimation, configuration management, project progress management. Use of Testopia for test case documentation and test management. Test Execution, logging defects, defect lifecycle, fixing / closing defects. Use of Bugzilla for logging and tracing defects

Unit IV Test Data Management 7

Test Data Management –Overview, Why Test Data Management, Test Data Types, and Need for Test Data Setup, Test Data Setup Stages, and Test data management Challenges. Creating sample test data using MS-Excel

Unit V Automation testing 11

Introduction to automation testing, why automation, what to automate, tools available for automation testing. Introduction to Selenium, using Selenium IDE for automation testing, using Selenium Web driver for automation testing, understanding TestNG framework with Selenium Web driver for automation testing

Total Periods: 45

Text Books:

S. No.	Author(s)	Title of the Book	Publisher	Year of Publication
1.	Rex Black	Managing the Testing Process (2nd edition)	John Wiley & Sons	2001
2	Dorothy Graham, Erik van Veenendaal, Isabel Evans, Rex Black	Foundations of software testing	Cengage	2018
3	Elfriede Dustin	Implementing Automated Software Testing: How to Save Time and Lower Costs While Raising Quality	Pearson	2009

Reference Books:

S. No.	Author(s)	Title of the Book	Publisher	Year of Publication
1.	M.G.Limaye	Software Testing – Principles, Techniques and Tools	Tata McGraw-Hill	2009
2.	Paul C. Jorgensen	Software Testing – A Craftsman's approach	Third edition, Auerbach Publications	2010.
3.	Rahul Shende	Software Automation Testing Tools for Beginners	Shroff Publishers and Distributors	2012

Web URL(s):

1. http://docs.seleniumhq.org/docs/
2. http://www.seleniumhq.org/download/